Kantonsschule Im Lee Winterthur

Maturitätsarbeit HS 2026/27

Zugängliche Informationen trotz Verschlüsselung, die Macht der Metadaten.

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Winterthur

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Quellenverzeichniss

[1] fhdrsdg. *How to expand a frame inside a canvas in tkinter*. 2018. URL: https://stackoverflow.com/questions/50243679/how-to-expand-a-frame-inside-a-canvas-in-tkinter (besucht am 10. 10. 2026).

[2] GeeksforGeeks. *Scrollable Frames in Tkinter*. 2026. URL: https://www.geeksforgeeks.org/python/scrollable-frames-in-tkinter/ (besucht am 10. 10. 2026).

[3] Peter Grill. *When to use \par and when \newline or blank lines?* URL: https://tex.stackexchange.com/questions/82664/when-to-use-par-and-when-newline-or-blank-lines (besucht am 13. 10. 2026).

[4] Freddy Mcloughlan. *Implementing biber for biblatex in Microsoft Visual Studio Code*. 2024. URL: https://tex.stackexchange.com/questions/459640/implementing-biber-for-biblatex-in-microsoft-visual-studio-code (besucht am 13. 10. 2026).

[5] Overleaf. *Creating a bibliography*. URL: https://docs.overleaf.com/citing-and-references/creating-a-bibliography (besucht am 13. 10. 2026).

[6] Overleaf. *How do I adjust the font size?* URL: https://www.overleaf.com/learn/latex/Questions/How_do_I_adjust_the_font_size%3F (besucht am 13. 10. 2026).

[7] Overleaf. *Learn LaTeX in 30 minutes*. URL: https://www.overleaf.com/learn/latex/Learn_LaTeX_in_30_minutes (besucht am 12. 10. 2026).

[8] Overleaf. *Page size and margins*. URL: https://www.overleaf.com/learn/latex/Page_size_and_margins (besucht am 13. 10. 2026).

[9] ron rothman. *Converting epoch time into the datetime*. 2023. URL: https://stackoverflow.com/questions/12400256/converting-epoch-time-into-the-datetime (besucht am 09. 10. 2026).

[10] TKDocs. *Tkinter Tutorial*. URL: https://tkdocs.com/tutorial/ (besucht am 23. 09. 2026).

[11] W3Schools. *Python Global Variables*. URL: https://www.w3schools.com/python/python_variables_global.asp (besucht am 07. 10. 2026).

[12] W3Schools. *Python Tuples*. URL: https://www.w3schools.com/python/python_tuples.asp (besucht am 04. 10. 2026).